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Atmospheres of alien worlds: Exploring new tools for high-resolution spectroscopy of exoplanet atmospheres

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Ultra-hot Jupiters

- Short orbital periods, close-in orbits and tidally locked
- Elevated dayside temperatures > 2400 K
- Atmospheres rich in volatiles + refractories
- Inflated atmospheres ideal for transmission spectroscopy (more light filters through)
- Constraining atmospheric composition is important for planetary formation theory and search for biomarkers

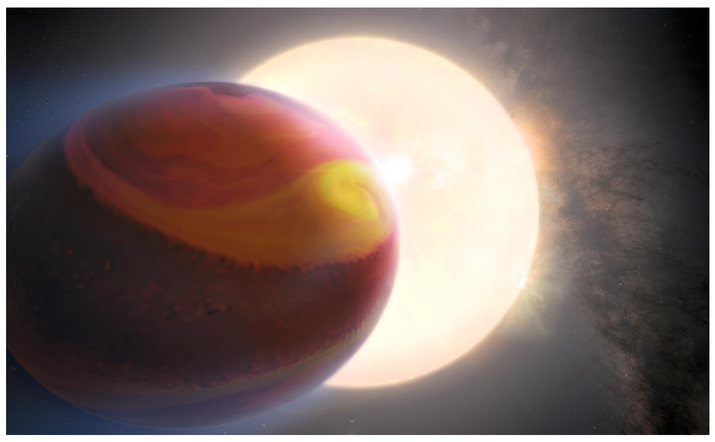


Figure 1: Artist's impression of WASP-121 b. An ultra-hot Jupiter orbiting close to its host star [1]

Isolating the Exoplanet Spectrum:

- Exoplanet signal needs to be untangled from overbearing radiance of host star
- Planet's radial velocity orders of magnitude greater than stars
- Planetary spectrum has a large Doppler shift relative to stellar background. Stellar spectrum quasi-static across time-series
- The large relative motion of the planet allows it to be separated from strong stationary features that dominate observation

Removal of Tellurics:

- High resolution time-series spectra of WASP-121 b observed with CRIRES+ spectrograph across K band (1.9 - 2.5 μm)
- Strong static absorption lines from Earth's atmosphere present in time-series spectra (vertical lines in Fig 3)
- De-trending algorithm SysRem removes static features that do not vary across time-series like telluric and stellar lines.
- Leaves a weak exoplanetary signal on a background of noise.

Cross Correlation with Model Templates:

- Exoplanetary spectrum is buried beneath photon noise
- Boost the exoplanetary signal with a predicted transmission spectrum of WASP-121 b via cross correlation
- The model template consists of the expected adsorption lines and needs to be Doppler shifted to match planet signal
- The spectrum and shifted model template are multiplied, and their product is summed over wavelength and spectral order

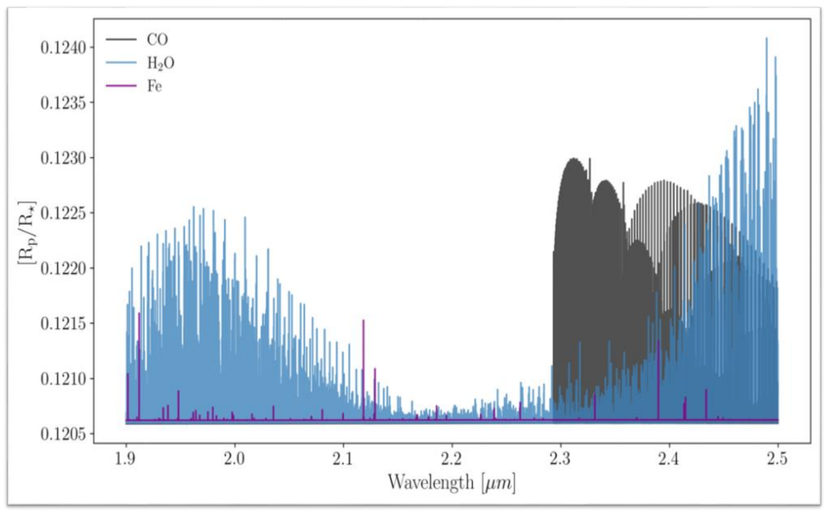


Figure 2: Modelled transmission spectra of species from Wasp-121b atmosphere. Templates created using petitRADTRANS[2]

CRIRES+ Observation → Cleaning, Alignment and Blaze Correction → Removal of Telluric Contamination → Cross Correlation with Model Template → Velocity Summation

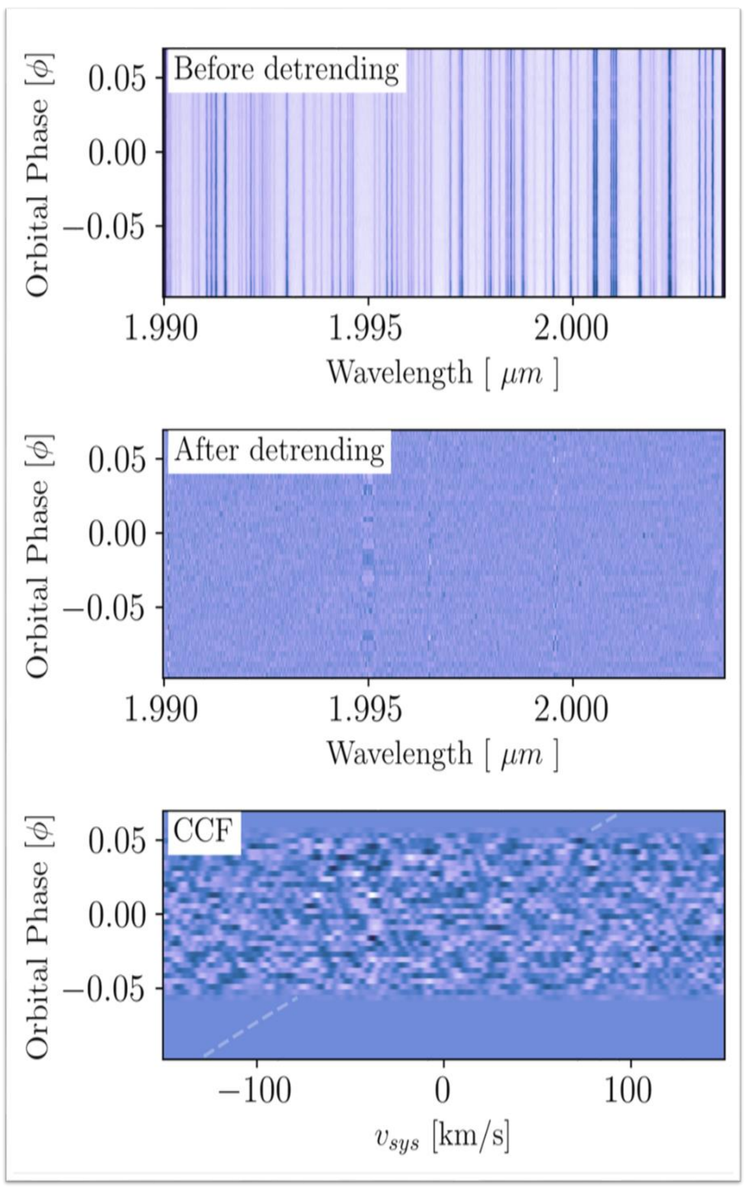


Figure 3: Spectrum after cleaning, blaze correction and alignment (Top) Removal of telluric and stellar features after dividing though by SysRem model (Middle). Summed cross correlation of spectra with CO model template (Bottom)

Velocity Summation:

- When working with weaker signals the cross-correlation time-series needs to be integrated over a range of predicted planetary radial velocities:

$$v_p = K_p \mathrm{Sin}(2\pi\Phi) - v_{sys}$$

- The change in the radial velocity of the planet leads to a clear feature for strong planetary signals, confidently confirming the presence of the species in the atmosphere
- Using pattern of molecular absorption features across a wide wavelength range and effectively summing up all those lines allows detection of even very weak signals
- Amplitude of the signal (detection significance σ) found by subtracting mean of an off-signal area and dividing through by standard deviation
- Method works to unambiguously confirm presence of species, CO detection shown to right! (little white blob)
- It is challenging to draw further information from signal due to loss of information about the continuum during data reduction. Requires more statistically robust measures to acquire abundances

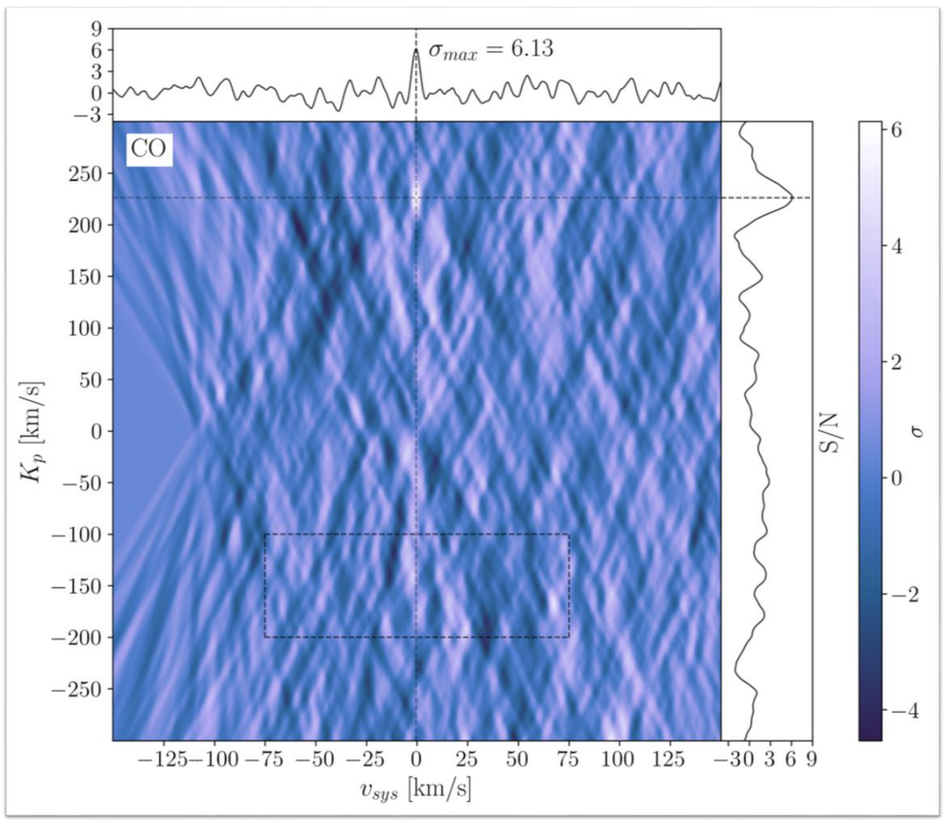


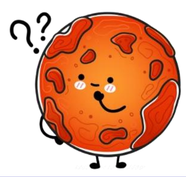
Figure 4: Cross Correlation Map of CO. Signal at v_{sys} = -0.45, K_p = 226 km/s. Dashed box indicates region used to calculate detection significance. Slices show noise levels through max signal.

Results:

- Clear detection of carbon monoxide with detection significance of 6.13 σ at expected planet position
- Non detection of H_2O indicating masking by strong CO feature, possibly also polluting CO detection strength
- Non detection of Fe due to weaker spectral feature in infrared
- Varied model template and number of SysRem iterations to maximise CO detection
- CO signal less sensitive to changes in modelled spectrum due to altering of signal during telluric removal
- More accurate constraints require future work!

Future Work:

- Full atmospheric retrieval and joint model templates to resolve masking of H_2O
- Simultaneous low-resolution + high resolution observation would allow both broad and narrow atmospheric features to be studied
- Joint optical and infrared observation would allow access to the full range of species in atmosphere



Citations:
[1] Credit: NASA, ESA, Q. Changeat et al., M. Zamani (ESA/Hubble)
[2] Molliére et al. (2019), Documentation available at https://petitradtrans.readthedocs.io

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